



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year: 2024 - 2025(Odd Semester)

Degree, Semester & Branch: B. Tech, V & IT

Course Code & Title: CS3551& Distributed Computing

Name of the Faculty member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III/ Suzuki-Kasami's Broadcast Algorithm
- **Course Outcome:** CO3
- **Activity Chosen:** Flipped class Room
- **Justification:**

A flipped classroom is a type of collaborative learning where students might watch videos took the notes according to their understanding level at home and then do the task provided in classroom with gained knowledge. This improves the habit of self-learning among the students.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

- Flipping the classroom is an inverting the classroom approach to teaching. In this approach, the traditional in-class teaching is "flipped" to better meet the needs of individual learners. Students gain control of the learning process through studying course material outside of class, using readings, pre-recorded video lectures. It helps the faculty/lecturer to redefine in-class activities and include homework problems and keep the students engaged in the content.

- **Identify and Share:** Related materials are identified and I posted relevant references/materials/notes of Suzuki-Kasami's Broadcast Algorithm to the students and given as self-learning.

- **Evaluate:** I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/present in class. The Figure.1 and Figure 2 images shows the student Joel Paulraj M and Thanishka Shree U explaining the self-explored concepts on Suzuki-Kasami's Broadcast Algorithm. Make the students to find out the answer by their own by learning.

- Images / Screenshot of the practice:



Fig 1: Joel Paulraj M presentation about Suzuki-Kasami's Broadcast Algorithm



Fig 1: Thanishka Shree U presentation about Suzuki-Kasami's Broadcast Algorithm

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO10	PSO1	PSO2
CO3	3	2	1	2	3	3

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:**

Innovative practice	PO1 3	PO2 2	PO3 1	PO10 2	PSO1 3	PSO2 3
Justification for correlation	Students will be able to use the basic mathematical knowledge, to solve Mutual exclusion and deadlock problem.	Students will be able to identify, formulate and analyze the best solution for problem with the knowledge of the Suzuki-Kasami's Broadcast Algorithm for Mutual exclusion.	Students could be able to provide the solution for complex problem with the knowledge of Mutual exclusion and deadlock concepts.	The students could be able to explain the Suzuki-Kasami's Broadcast Algorithm.	Students could apply the knowledge of features supported by various UI design in developing reliable IT solutions.	Students can be able to apply appreciable knowledge of design and development practices to deliver a quality outcome in software project.

Reflective Critique:❖ **Feedback of practice from students and other stakeholders:**

- Student felt good, since they can study at their own pace/time.
- They felt that through such learning, they can explore more.

❖ **Benefit of the practice:**

- More one-to-one time between teacher and student.
- More collaboration time for students.
- Students learn at their own pace.
- Practical things – like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

❖ **Challenges faced in implementation:**

- Relies on student preparation – few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped classmaterial is higher than for a traditional class.

References:

- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162- what-why-and-how-to-implement-a-flipped-classroom-model](https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model)
- ❖ <https://educationaltechnologyjournal.springeropen.com/articles/10.1186/s41239-016-0032-z>

Signature of Faculty Member

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